



# United International University (UIU)

## Dept. of Computer Science & Engineering (CSE)

### Final Exam: : Trimester: Summer - 2024

Course Code: CSE 4889, Course Title: Machine Learning, Sec: A

Total Marks: 50

Duration: 2 hours

**Please answer all the questions.** Figures are in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

#### Question 1:

5+5=10

- A. Explain the steps of k-Means Clustering algorithm and its disadvantages.
- B. Cluster the following eight points (with (x, y) representing locations) into three clusters:  
A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9)  
Initial cluster centers are: A1(2, 10), A4(5, 8) and A7(1, 2).

The Euclidean distance is used between each point and the centroid of its associated cluster.

#### Question 2:

5+5=10

- A. What is ensemble clustering? What are the three ensemble clustering methods?
- B. How can Clustering (Unsupervised Learning) be used to improve the accuracy of the Linear Regression model (Supervised Learning)? Explain with an example.

#### Question 3:

4+3+3=10

- A. What are some common challenges found in deep learning models, and how would you overcome them?
- B. How would you evaluate the performance of a deep learning model?
- C. How would you design and implement a deep learning model for a problem with limited labeled data?

#### Question 4:

5+5=10

- A. What is backpropagation? How does backpropagation work? Explain with an example.
- B. Apply the gradient descent technique to find the new intercept from Table 1. Where learning rate is 0.1, initial intercept value is 0 and slope is 0.64.

Table 1

| Instance       | X1  | y   |
|----------------|-----|-----|
| x <sub>1</sub> | 0.5 | 1.4 |
| x <sub>2</sub> | 2.3 | 1.9 |
| x <sub>3</sub> | 2.9 | 3.2 |

#### Question 5:

5+5=10

- A. What is feature mapping and padding in CNNs? Why do we need padding?
- B. Explain the core components of Convolutional Neural Network with examples.